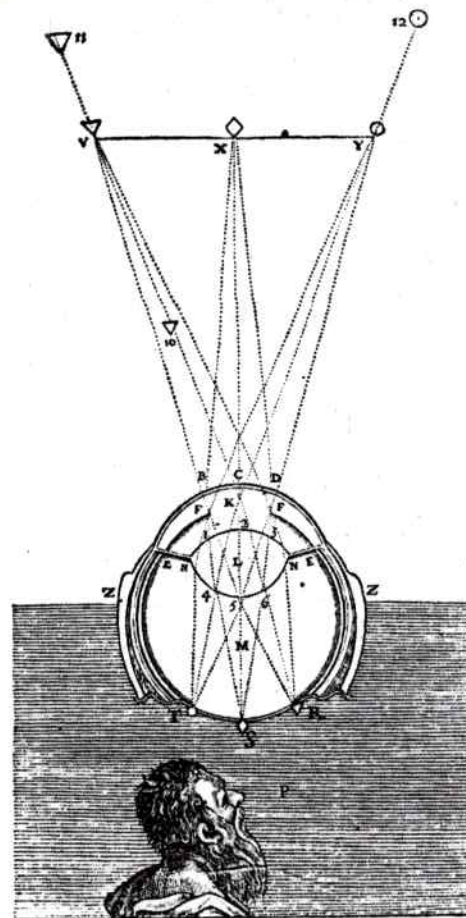


Colours in the Eye of the Beholder



↑ Illustration of the theory of the retinal image. In Descartes, René. *La Dioptrique*. Leiden, 1637.

The colours we can see are actually receptors in our eye receiving different wavelengths of light. The brain interprets the different wavelengths as hues of colour. We have three different receptors for colour called cones: the short wavelength cone is called a blue cone; the medium wavelength is called the green cone; and the long wavelength is called red cone. These receptors are responsible for all the names and categories of colour that are found in history.

The human eye can theoretically distinguish between 7.5 and 10 million hues.¹ About 8 percent of men and 0.5 percent of women have some type of colour blindness. The two major types of colour blindness are those unable to distinguish red and green (Deuteranopia and Protanopia), and those unable to distinguish green and blue (Tritanopia). Colour blindness is nearly a hundred times more likely amongst men than women because it is inherited through the X-chromosome. Colour blindness is much more common than often assumed, so make sure designs do not rely on colour discrimination alone.

RED
ORANGE
YELLOW
GREEN
BLUE
PURPLE

Deuteranopia
8.1% of men
0.36% of women

RED
ORANGE
YELLOW
GREEN
BLUE
PURPLE

Protanopia
2.45% of men
0.04% of women

RED
ORANGE
YELLOW
GREEN
BLUE
PURPLE

Tritanopia
0.011% of men
0.04% of women

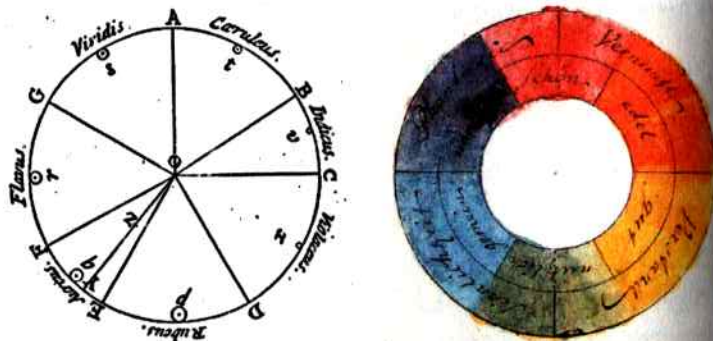
The wavelengths of light human beings can sense is about 380nm to 720nm.² How colours in nature are perceived all depend on the viewer. Humans have three colour receptors, but some animals have four or even six colour receptors, and can see colours that cannot be seen by humans. Some animals, like bees, have an extra UV receptor that is sensitive to ultraviolet light.

1. Novin, Guity. 'A History of Color, Color Wheel, and Psychological Impact of Color'. *A History of Graphic Design*. June, 2014. guity-novin.blogspot.nl/2014/07/chapter-70-history-of-color-color-wheel.html.

2. Holtzschue, Linda. *Understanding Color*. John Wiley & Sons, Inc., 2011.

Colour Systems

Sir Isaac Newton was the first who used a triangular prism in 1676 to break white sunlight into different colours of light. He proved that colour is part of light itself, something that had been debated for years. Newton first used the word 'spectrum' for the array of colours that the prism produced. From Aristotle in 550 B.C. to the present-day, scientists and artists have tried to describe and organise the colour spectrum that is visible to humans into some kind of system of harmony. Although we have many practical uses in the industry for colour systems like CMYK and RGB, discriminating between different colours and creating categories of colour are in fact very culturally dependent.



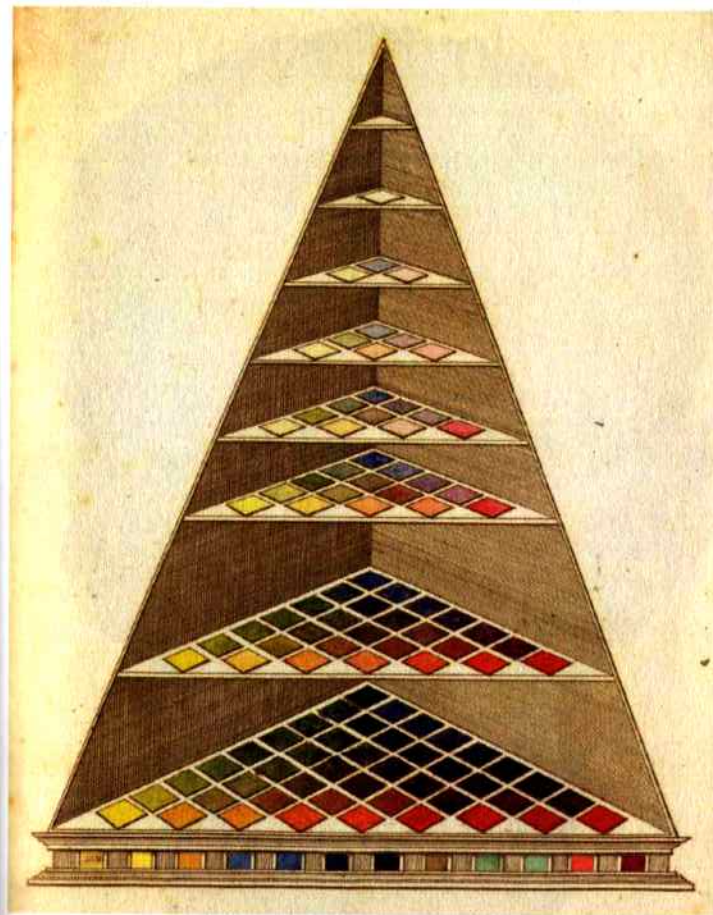
The Mayans and Incas of Mesoamerica had colour categories which were unknown in the rest of the world. The Mayans had a name for a specific colour in between green-blue and green, which was used to name objects incrustated with a mosaic of jade and turquoise.¹

The Inca language *Quechua* had a colour name for the combination of black and white, which suggest a relation with the many black and white checkerboard patterns found in Inca textiles.¹

↑ Newton's colour wheel, in: Newton, Isaac. *Opticks*. 1704, from Book I, Part II, Proposition VI, 1704.
 ↗ Goethe's colour wheel, in: Goethe, Johann Wolfgang von, *Zur Farbenlehre*, 1810.

1. Gage, John. *Color and Meaning: Art, Science, and Symbolism*. University of California Press 1999, pp. 106-107.

2. Gage, p. 30.



The steadfastness of some Europeans who tried to find universal truth in colour systems led to some unexpected encounters. In 1971 two Danish anthropologists arrived on Bellona island in Polynesia armed with a book of colour swatches, when they were told by an islander, 'We don't talk about colour here'.² The belief that there are universal systems of colour shared by all cultures, is a romantic idea that originated in nineteenth century Western Europe.

↑ Lambert, J.H. *Beschreibung einer mit dem Calauschen Wachse ausgemalten Farbenpyramide*. Haude und Spener, 1772.



For centuries, the standards of beauty were the pale marble and bronze sculptures of ancient Greece and Rome. It turns out the quintessence of aesthetics for Western sculpture was based on a misconception.

↑ 'Caligula' (37–41 CE), marble, h: 28 cm, Ny Carlsberg Glyptotek, Copenhagen, Denmark, inv.no. IN 2687.



In the 2000s, scientists proved these sculptures were actually painted in bright colours, by scanning them using infrared, X-Rays, and UV light. Read more about this research at www.trackingcolour.com.

↑ 'Caligula' (reconstruction), 37–41 CE (2011), marble, h: 28 cm, Archäologisches Institut der Universität Göttingen and Stiftung Archäologie, Munich, inv.no. IN 2687.

The Cost of Colour



Before the mid-nineteenth century colour pigments were made from minerals, animals, and plants. Making natural dyes was a costly and time-consuming task. The brightest pigments, like ultramarine, gold, and vermillion, were only available to the richest patrons.

Gold paint was made from real gold that was ground up into a powder and mixed with arabic gum. Goldleaf was made by beating a piece of gold into an ultrathin layer of less than a half a millimetre. Ultramarine was one of the most expensive and desired colours. This bright blue pigment was made from the hard lapis lazuli stone, which at the time could only be mined in present-day Afghanistan. Owning paintings in these colours was a sign of tremendous wealth.

↑ *Virgin Mary*, painted by Giovanni Battista Salvi Di Sassoferrato between 1640–1650 with ultramarine. National Gallery, London.

In pre-Columbian Mexico, the Aztecs and Mayans produced a very special red dye called *cochineal*. It was made by collecting insects from the nopal cactus, which was a time-consuming process. The Aztecs and Mayans used it to paint their temples bright red. When the Spanish invaded, they started exporting the pigment to Europe. Soon after, cochineal became Spain's second-most valued export from Mexico after silver. In the late nineteenth century, the invention of synthetic dyes significantly lowered the cost of pigments. Cochineal is still used today as a natural dye for food and cosmetics.

Though they might not have the allure of gold paint or ultramarine, some colours are still more costly than others. Fluorescent or glow-in-the-dark inks require a higher pigmentation. Gold and silver inks contain metallic particles, making those inks more opaque than standard transparent printing inks. The brightest inks are still the most expensive.



↑ The Aztec god Quetzalcoatl (feathered serpent). Coloured with cochineal, Codex Magliabechiano, mid-sixteenth century, Mexico.

Drop Food, Not Bombs

They are both yellow, but one contains a 2000 calorie meal and the other is capable of killing anyone within a 50-meter radius. In 2001 the U.S. dropped cluster bombs on Afghanistan in its war against the Taliban. One in ten cluster bomb units failed to explode, posing a renewed threat to those on the ground. At the same time, food rations were dropped over Afghanistan, in packaging of the same colour and the same typography as the bomb units. In November 2001 the U.S. government realised the mishap, and changed the colour of the rations to pink. Leaflets were made to explain the difference. More than a million food rations had been dropped by then.¹



↑ Humanitarian Daily Ration (HDR). Photo: Bill Dugan.

↗ BLU-97/B cluster bomb unit. Photo: Adam Newsome.

FRONT



BACK



↑ Leaflets warning how to tell the difference between bombs and food package. Image: CNN.
1. 'U.S. Changes Color of Food Aid.' CNN. November 1, 2001. edition.cnn.com/2001/US/11/01/ret.afghan.fooddrops.

Stop for Yellow

In 1924, before stops sign were red, a yellow version was used in the U.S. This was before reflective red paints were available, and yellow had a better visibility than red, especially at night. In 1954 the stop signs in the U.S. were all painted in a reflective red.



Colour Psychology

How people respond to colour is of great interest to those who work in marketing. Colour psychology research is often focused on how the colour of a logo or a product will yield higher sales, and what colour preferences can be found in certain age groups and cultures.

The study of the psychological effects of colour have coincided with colour theory in general. Goethe focused on the experience of colour in his *Zur farbenlehre* from 1810,¹ in opposition to Sir Isaac Newton's rational approach. Goethe and Schiller coupled colours to character traits: red for beautiful, yellow for good, green for useful, and blue for common. Gestalt psychology in the early 1900s also attributed universal emotions to colours, a theory that was taught to students at the Bauhaus by Wassily Kandinsky.

The best known colour psychology test was conducted by Max Lüscher in the 1940s, when he tested the responses of people to a series of coloured cards. A psychologist could interpret the person's character based on the test results. The interpretation of this research in colour psychology is disputed, since it is not clear whether people respond to the colours or the cultural concept of the colour. It is more likely that the cultural context, our upbringing, and personal preferences influence our interpretation of colour more than anything else.²

Some colour psychology findings have been proven to work. 'Cooler' colours can have a calming influence on people, which is why hospitals and prison walls are often painted in a soft greenish colour. Another research shows that the colour of placebo pills influences their effectiveness. Blue coloured pills work better as depressants, and red colour pills better as stimulants.³

1. Goethe, Johann Wolfgang von, *Zur Farbenlehre*, 1810.

2. Gage, John. *Color and Meaning: Art, Science, and Symbolism*. Berkeley: University of California Press, 1999, p. 31.

3. Cohen, Tessa Fiorini. 'The Power of Drug Color.' *The Atlantic*. October 13, 2014. www.theatlantic.com/health/archive/2014/10/the-power-of-drug-color/381156.

Colour and Culture

Research shows that ninety-eight languages have words for the same eleven basic colours;⁴ however, the meaning a colour may have can be very different. There are conflicting theories on whether the cultural meanings of colours can be categorised. Meanings can change over time and depend on the context. Black may be the colour of mourning in many countries, though a black book cover or a black poster is not always associated with death. Another example is that brides in China traditionally wear red, but many brides have started to wear white in recent decades. The cultural meaning of colours is not set but always changing. The next few pages list some of the meanings of colours in different cultures.

Black

was the colour of mourning during the Roman empire, and in Judaism and Christianity it is associated with death and evil. During the Middle Ages, black clothing became popular, and during the Renaissance it was worn by the wealthy and nobility in all parts of Europe. In the nineteenth century, the black army was the nickname of the Russian anarchists. In the twentieth century, black became the colour of Italian and German fascism, who were called blackshirts. In Shia Islam black is a very devout colour. The black standard is one of the flags of Muhammad in Islam. The black standard has been appropriated as a symbol by jihadism since the late 1990s.

White

is the colour of death and mourning in China, Vietnam, and Korea. In India, people wear white after the death of a family member. Jewish, Christian and Hindu religions use the colour white in rituals to indicate purity, chastity, virginity, and peace. White is the traditional colour of bridal dresses in Western cultures. In Europe, white was historically associated with absolute monarchy. It became synonymous in the twentieth century with anti-communist and counter-revolutionary movements in Russia. A white dove is an international sign of peace, and a white flag is the international sign of surrender.

Yellow

is the most visible colour from a distance, so it is often used for objects that need high visibility, such as ambulances, road maintenance equipment, and taxis. In East Asia, yellow is considered sacred and imperial. In China it is the colour for royalty, and during the Qing Dynasty, only members of the imperial household were permitted to wear yellow. In China it is still seen as the colour of happiness, glory, wisdom, harmony, and culture. Yellow is used to symbolise gold as one of the colours of the Catholic church and the Vatican. Yellow in Italian is *giallo* and a nickname for crime stories. The demonstrations in the Philippines in 1986 were also known as the yellow revolution.

Orange

comes from the Sanskrit word for orange tree. In Asia it is known as saffron, named after the spice. In Hinduism, the divinity Krishna is portrayed dressed in saffron. In Buddhism, saffron is the colour of illumination, the highest state of perfection, and the colour of robes worn by monks. Orange ribbons are used to promote awareness and prevention of self-injury. In Northern Ireland the protestant order is called the Orangemen, who wear orange during annual marches. Orange is the colour of the Dutch monarchy, and in South Africa, orange was the colour of the ruling party during Apartheid. Prisoners are often dressed in orange to make them distinctive in case of escape.

Red

comes from the Sanskrit word for blood. In Greek mythology it was the colour for the planet Mars and the god of war. Red is the most important colour in China and means good luck and happiness. Red promotes long life, and it is used in Chinese new year celebrations. Red is the traditional colour for brides in India and Nepal. In Japan, a red kimono symbolises happiness and good luck. In Japan's Shinto religion, red is the symbol of life. The Russian word for red has the same root in Old Russian as the word 'beautiful'. The oldest symbol of socialism is the red flag, and red is still associated with socialist and communist parties. The red shirts, or the red movement, was the name of those who opposed the 2006 military coup in Thailand.

Green

is the colour of nature in many cultures. Early rituals centred on the hope of a good harvest with green vegetables. Green is the traditional colour of Islam, the colour of the prophet Muhammed, which is why many flags in the Islamic world use the colour green. Since the 1980s, green has become the colour of environmental parties and organisations. Green parties in Europe have programmes based on ecology and environment. The word greenwashing is used to describe the advertising of companies that use positive environmental practices to cover up environmentally unfriendly activities. A 'green room' is used in television and theatre to quiet nervous performers.

Blue

is tested as the colour that is best liked by all cultures, which is how it became the colour for international business. The United Nations chose blue as a colour in 1946, but the colour has no special significance. Bluewashing is a term for companies that join the United Nations 'Global Impact Initiative' only for PR reasons. 'Blue collar worker' refers to the shirts worn by people working manual labour jobs. The Tuareg of North Africa wear blue turbans, dyed with indigo. Azure blue is the national colour of Italy, the national soccer team is called the *Azzurri*. Several languages, like Japanese, Thai, Korean, and Lakota Sioux, use the same word to describe blue and green.

Purple

was the the most expensive dye in Roman times, and was worn by Roman magistrates, Byzantine emperors, and the rulers of the Holy Roman Empire. During the reign of Elizabeth I, only royalty was allowed to wear purple in England. Roman Catholic bishops still wear purple as a colour of royalty. In Japan, the colour is traditionally associated with the emperor and aristocracy. In Thailand, widows in mourning wear purple. In Western politics the colour purple is neutral, and used for a coalition between a left-wing and a right-wing party. The Women's Suffrage movement used purple, and the colour was adopted in the 1970s by the women's liberation movement. Purple and pink are both associated with the LGBT community.



21:54, 28 September 2005 by Zscout370



08:00, 5 February 2006 by Denelson83



21:33, 12 February 2006 by Vzb83



13:36, 23 September 2008 by Flad



01:45, 27 September 2008 by MaggotMaster



23:46, 27 September 2008 by Flad



22:01, 9 October 2008 by Avala



19:02, 10 October 2008 by Flad



21:43, 10 October 2008 by Reioio



21:08, 26 September 2011 by Zscout370

National Colours

With Yuri Veerman

The flag has been the most prominent symbol of national identity since the nineteenth century. A new national flag often marks a revolution or a political change. Flags have a military origin, and were used to identify military units from one another in the Roman legions.

Different national flags often use similar colours, sometimes even with the same meaning. Thirty-eight percent of the world's national flags use red as a colour, which symbolises the blood spilt for independence, or on the battlefield. One of the most used colour combinations is red, white, and blue (Chile, Panama, Russia, Samoa, United Kingdom, Thailand, etc.) where blue often symbolises the domination over sky or sea.¹

Many African countries use red, black, and green in their flags. This stems from the Pan-African flag from 1920.² The red symbolises the blood spilled for liberation, black is for the African people, and green for the continent's natural wealth.

There have been a few monochrome flags. Morocco had a deep red flag for over two hundred years. Afghanistan had an all black flag at the end of the nineteenth century. And during the reign of the Taliban in 1996, Afghanistan had an all white flag. The last remaining monochrome flag in the world was the Libyan flag, a plain green field without markings or insignia.

The green flag was a personal choice of Muammar Gaddafi whose manifesto was called the 'Green Book'. Artist Yuri Veerman collected the different digital versions of the Libyan flag that were discussed on the English Wikipedia page, 'Flag of Libya'. Between 2005 and 2011 the chromatic representation of Libya was revised fifteen times, each with different shade of green. After a six-year fight over the right shade of 'Gadaffi green', a fifteenth revision was made, marking the fall of the Libyan regime and the last monochrome flag.

© Yuri Veerman, 'Libya in Transition', 2012.

1. Byrne, Marie. 'Culture & Communications: Similarities of Color Meanings Among Diverse Cultures' *The Journal of Global Issues & Solutions*. BWW Society, 2003.
2. 'UNIA Declaration of Rights of the Negro Peoples of the World', New York, August 13, 1920. Reprinted in Robert Hill, ed., *The Marcus Garvey and Universal Negro Improvement Papers*, vol. 2, Berkeley, University of California Press, 1983.

Colour and Gender

Why is pink the colour for girls and blue the colour for boys? In the nineteenth century, girls and boys were both dressed in white because it was easier to clean. After World War I, department stores in the U.S. realised they could make more money if baby products would be gender specific. Pastels were in fashion, and it was decided that pink would be the colour for boys, and blue for girls.

In the 1940s, market research in the U.S. suggested the colours should be the other way around, and the 'baby boom' generation was the first where the girls were dressed in pink and the boys in blue.'



Children are not gender aware until age 6 or 7.² Exposing them to visual stereotypes at an early age creates an artificial view of gender. Now that parents can know the child's gender before birth, gender specific baby items have increased in popularity. Parents stock up on baby merchandise even before the baby is born. Jeongmee Yoon wrote her thesis on colour and gender, and since then has photographed children around the world with their pink and blue things.

↑ Sehyun and Her Pink Things, 2007. ↓ Jake and His Blue Things, 2006. Jeongmee Yoon, *The Pink Project*, 2005-ongoing.

1. Maglaty, Jeanne. 'When Did Girls Start Wearing Pink?' *Smithsonian Magazine*. April 7, 2011. www.smithsonianmag.com/arts-culture/when-did-girls-start-wearing-pink-1370097.
2. Halim, May Ling and Diane Ruble. 'Gender Identity and Stereotyping in Early and Middle Childhood' in: *Handbook of Gender Research in Psychology*, Springer, 2010, pp. 495-525.



AIRPORT SECURITY CHECK



The History of Colour in Black and White

'Look in your dictionaries and see the synonyms of the word Black. It's always something degrading and low and sinister. Look at the word White, it's always something pure, high and clean.'—Martin Luther King Jr., 1968.¹

In colour theory, white is the colour that includes all colours, while black indicates the absence of light. White is the standard background colour for drawing or designing. In printing white is without colour, while black is the darkest possible colour. Black absorbs all light, equalling darkness and shadow. This opposition can be found throughout the history of colour symbolism.²

In Christian and Jewish religions, white is synonymous with cleanliness, light, safety, and the good. White is used in opposition to black, both symbolically and literally. Black is dirty, and signifies danger and darkness. Angels are depicted as white, demons are black.³

How does this relate to racism? There is not one person who actually has a white or a black skin. Skin tones are variations of brown and pink hues. Europeans started calling themselves white around the same time that Africans were being sold as slaves.⁴ Categorizing Africans as black and Europeans as white, established the racial hierarchy that was needed to legitimise exploitation and slavery.

Today we can find the same symbolic opposition occurring in visual culture. Cinderella has light hair, and her sisters and stepmother have dark hair. In Star Wars, Luke Skywalker is the light-haired hero with the white skin, while his evil opponent Darth Vader wears black.² Although this might happen unconsciously, good and evil as a binary opposition between black and white still inhabits every corner of our visual language. Affirming these stereotypes in visual communication would be a continuation of colonialist ideas of race relations.

Image: i.imgur.com. Meme based on *Family Guy*. Created by Seth MacFarlane. 20th Century Fox Television.

1. Martin Luther King Jr. Speech on April 3, 1968.

2. Dyer, Richard. *White*. Routledge, 1997.

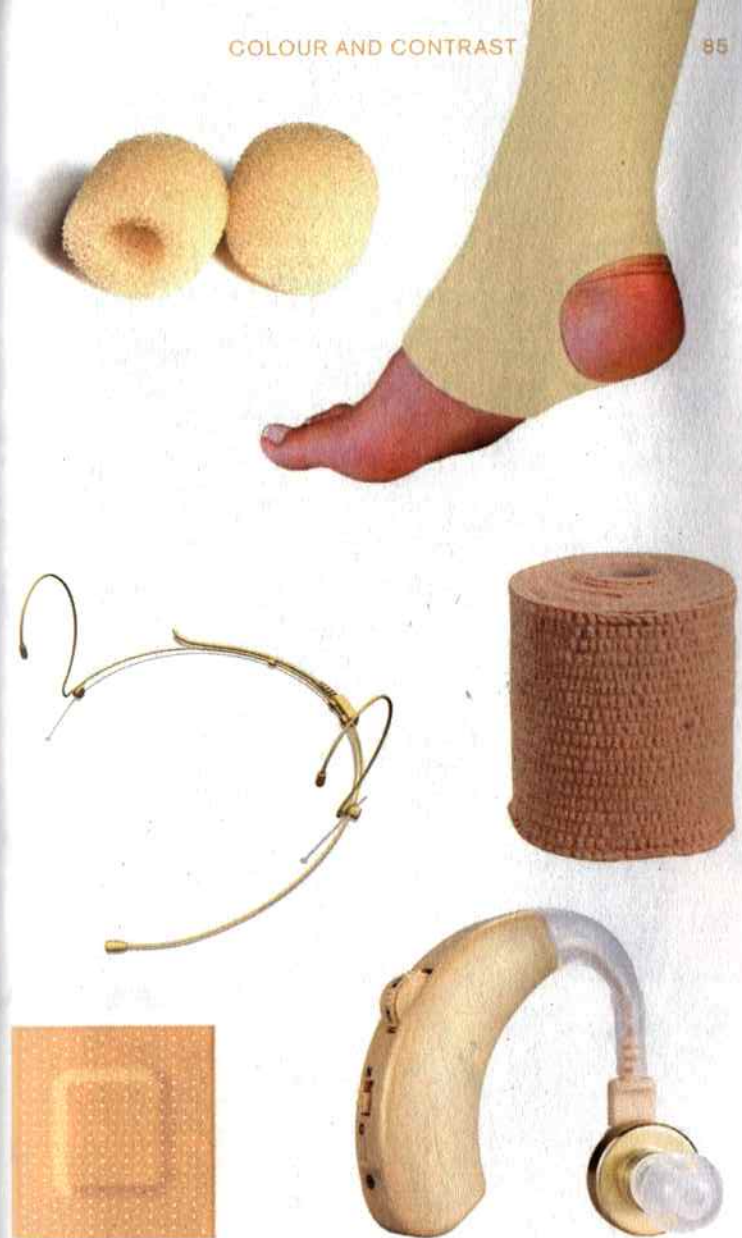
3. Hanssen, Eirik Frisvold. *Early Discourses on Colour and Cinema*, Stockholm Cinema Studies, Almqvist & Wiksell, Stockholm, 2006, pp. 62-73.

4. Ambrose, Stephanie. *Black: The Color of Language*. Massachusetts Daily Collegian. March 27, 2012. dailycollegian.com/2012/03/27/black-the-color-of-language.

'Research—into books, museums, the press, advertising, films, television, software—repeatedly shows that in Western representation whites are overwhelmingly and disproportionately predominant, have the central and elaborated roles, and above all are placed as the norm, the ordinary, the standard.'

— Richard Dyer
in *White*, 1997.

1. Dyer, Richard. *White*. Routledge, 1997.



To Colour or Not to Colour

Is there such a thing as being afraid of colour? The movie industry certainly had its moment of fear. Colour had been available since 1915, but was not widely adopted until the 1960s, after television had first switched to colour. Using colour film was enormously expensive, but more importantly, it was not regarded as being of the same artistic value as black and white. It was believed colour would distract people from the story, and colour film would not be able to show the same colours on screen as people saw in nature.¹

In photography and advertising, colour correction is routinely used to intensify colours, making images more striking. Colour correction can also challenge the legitimacy of the image. We still hold the belief that colours should be shown as naturally as possible.

Palestinian architect and photographer Yazan Al-Khalili uses colour correction to change the perception of Palestine in the media. On the left is his image of refugee camp Al-Amari close to Ramallah city. Palestinian refugee camps in the media are always depicted as landscapes of colourless damaged concrete.

Al-Khalili: 'Changing the refugee camp's colours is a symbolic act to fill the loss—like a child filling a colouring book—and produce the possibility of hope. Here I'm attempting to appropriate an urban landscape that reminds us of the tragedy—of their existence and our disappearance—in order to subvert memory into a desired future'.²

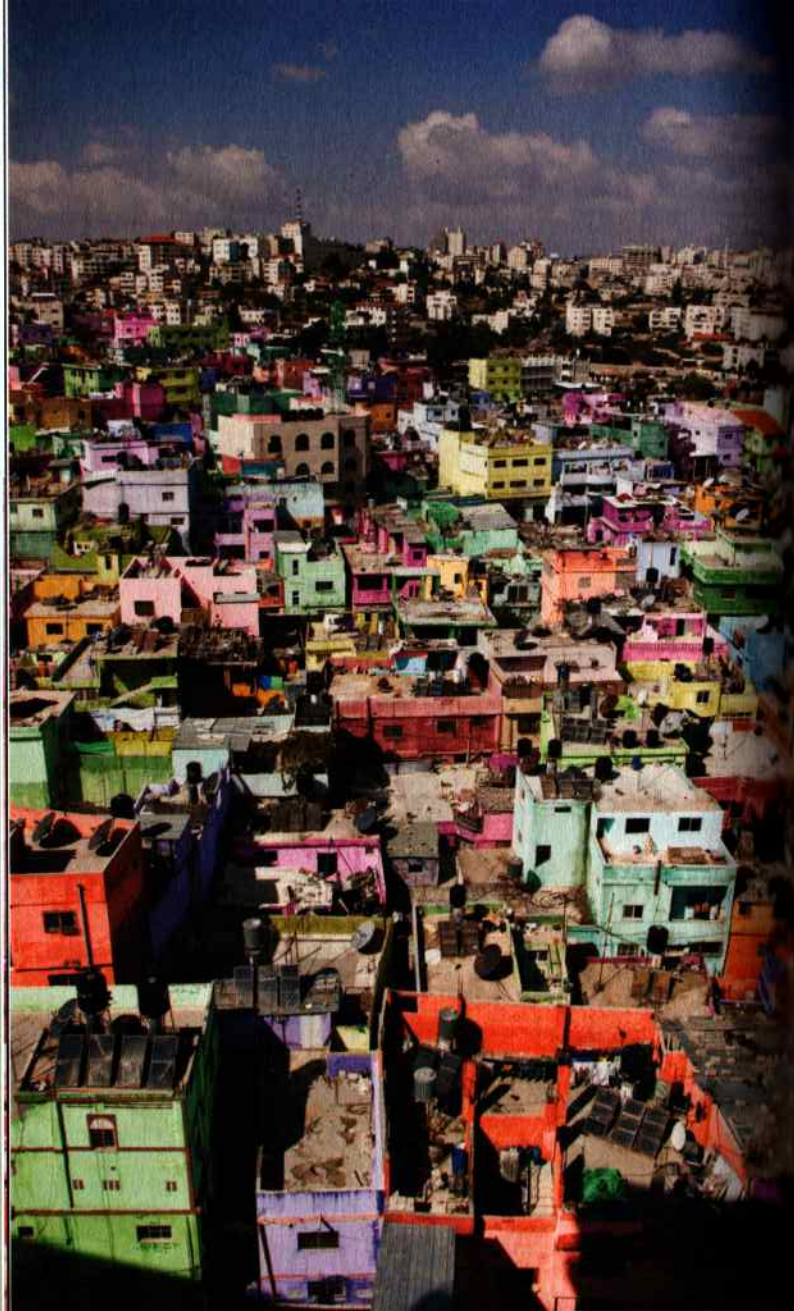
The act of colouring can also play on social conventions and perception. Ryan Hunter and Taige Jensen from New York make colouring books for grown-ups. Hunter and Jensen: 'Colouring for Grown-ups prepares you for the grim realities of adulthood in all the ways that the activity books of your youth irresponsibly did not. Proving that in a world that is too often black and white, or red and blue, sometimes you just need to add a few extra colours to your palette'.³

← Yazan Al-Khalili, *Colour Correction II - Camp Series*, 2007 - 2010.

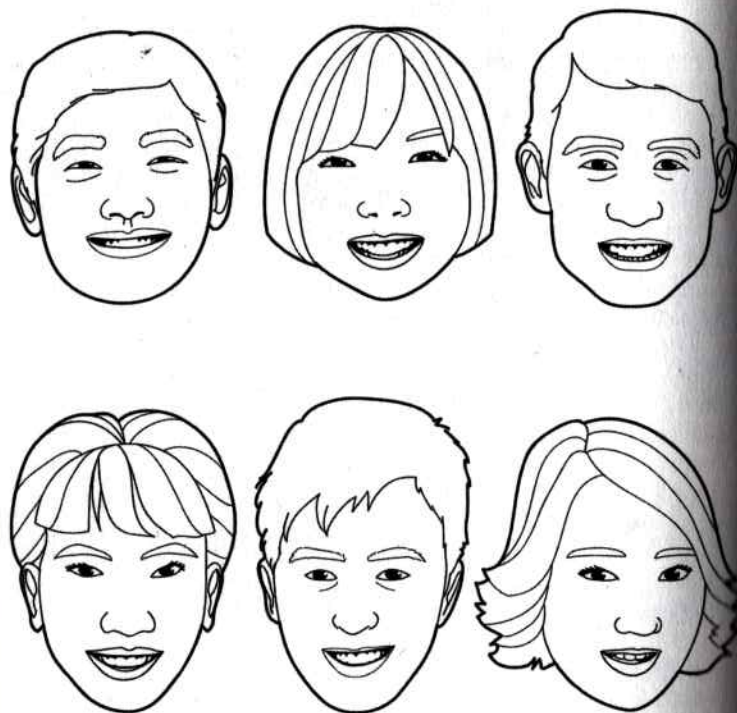
1. Elfassy Bitoun, Rachel. 'A History of Colour: The Difficult Transition from Black and White Cinematography.' *The Artifice*. April 21, 2015. the-artifice.com/history-of-colour-film.

2. Artist text from yazankhalili.com/index.php/project/colour-correction---camp-series.

3. Hunter, Ryan, and Taige Jensen. 'Coloring For Grown-Ups.' *The Huffington Post*. December 7, 2012. www.huffingtonpost.com/ryan-hunter/coloring-grown-ups_b_2251177.html.



CELEBRATE CHINESE NEW YEAR BY COLOURING THE CHINESE PERSON



**(Remember, there are Koreans, Japanese,
Filipinos, and other Asian races on this page!)**
Only colour the Chinese person!

↑ Ryan Hunter and Taige Jensen. *Coloring for Grown-Ups*. 2012-2015.

HIPSTER or HOMELESS?

**USE MAGIC MARKERS TO MAKE THE MAN ON
THE RIGHT INVISIBLE TO SOCIETY!**



SECRET HINT: While both hipsters and hobos share a mutual appreciation for beards, old clothing, and dirt, most hipsters give themselves away via their unwavering commitment to colour-coordination

Shirley and the China Girls

It is not clear what happened to Shirley. What we do know is that she worked at Kodak in the 1950s and she was the first model to pose for a Kodak test card. These cards were used as a standard to calibrate film colours with skin tones and colour bars. After Shirley, other women took her place and the cards became known as 'Shirley cards'.¹ The women needed to have a light skin, 'as smooth as "china"', hence the name 'china girls' for the women who worked in the movie industry and appeared on test cards for movies.

The Kodak films used in the 1950s, '60s, and '70s were calibrated for people with a white skin, and unsuitable to depict different races together. If you wanted to have a black person and a white person together on the same image, one of them would be either too light or too dark.² French director Jean-Luc Godard refused to use Kodak film in 1977 when he was commissioned to make a movie in Mozambique, because according to him the film was 'racist'.²

In 2012, photographers Adam Broomberg and Oliver Chanarin wanted to examine the 'the radical notion that prejudice might be inherent in the medium of photography itself' in their project *To Photograph the Details of a Dark Horse in Low Light*.³

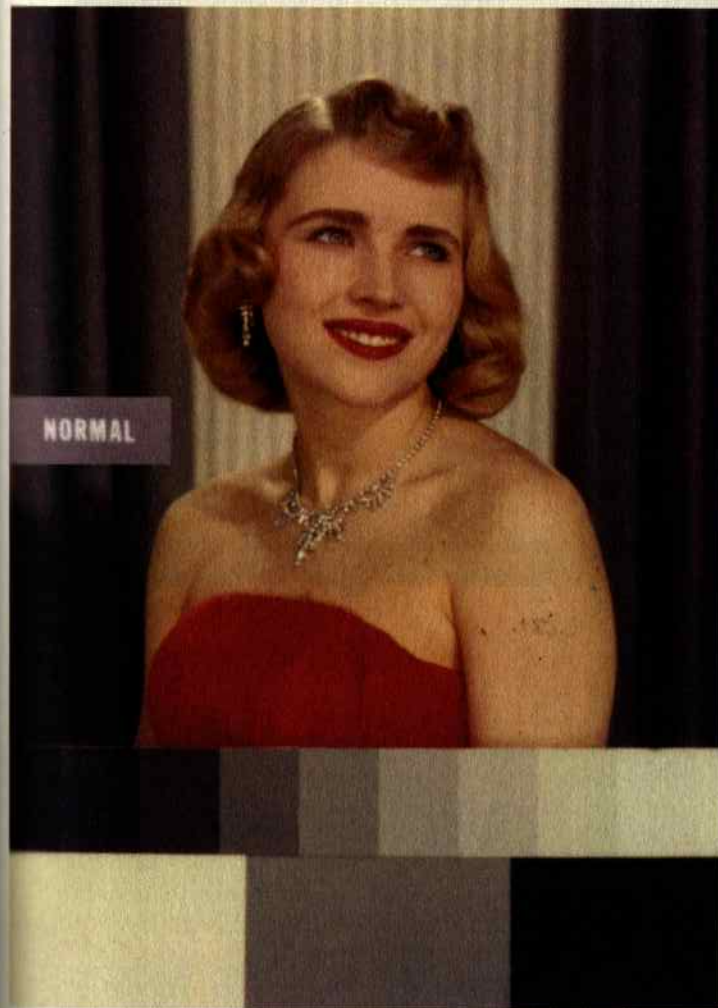
Test images are designed to represent a default, a set of standards for optimal image analysis. But tools are never completely neutral, and their settings reflect the cultural bias of the technicians who calibrate them. In their work, Broomberg and Chanarin show how the tools that are used for image production are not without built-in prejudices.

By the 1980s, Kodak films were configured to depict darker tones more appropriately. Not because of complaints of racism, but because clients had complained that chocolate and dark horses appeared too dark in the photos.²

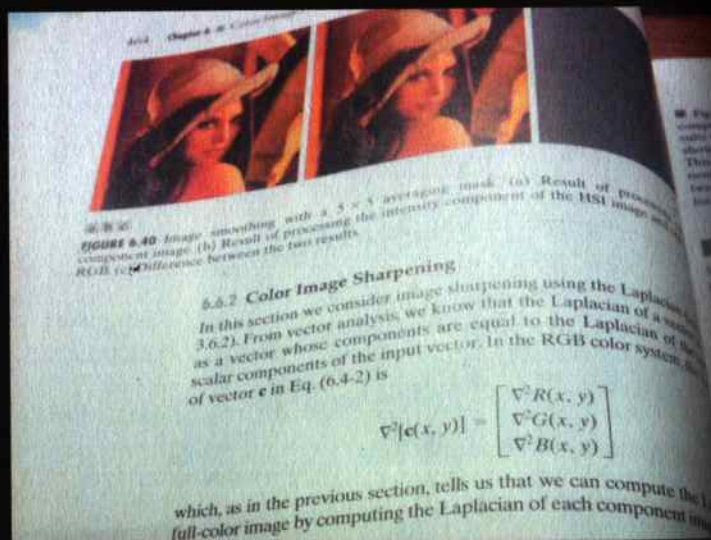
1. Del Barco, Mandalit. 'How Kodak's Shirley Cards Set Photography's Skin-Tone Standard.' *NPR*, November 13, 2014. www.npr.org/2014/11/13/363517842/for-decades-kodak-s-shirley-cards-set-photography-s-skin-tone-standard.

2. Smith, David. "Racism" of Early Colour Photography Explored in Art Exhibition.' *The Guardian*, January 25, 2013. www.theguardian.com/artanddesign/2013/jan/25/racism-colour-photography-exhibition.

3. O'Toole, Sean. *Yes, But (Some Thoughts on Broomberg and Chanarin's New Work)*. Toronto: Gallery TPW, 2013. gallerytpw.ca/rd/yes-but/



† *To Photograph the Details of a Dark Horse in Low Light (Shirley 1)*, Adam Broomberg & Oliver Chanarin, 2012. Courtesy of the Artists and Lisson Gallery, London.



† Photo of textbook *Digital Image Processing Second Edition*.

† Image: Ana Huaman, *Cascade Classifier*, Division of Natural Sciences and Engineering Swarthmore college, 2014.

1. Gonzalez, Rafael C. and Richard E. Woods, *Digital Image Processing Second Edition* Prentice-Hall, 2002.

2. Merchant, Brian, 'Lenna, The First Lady of The Internet.' *Motherboard*, August 20, 2011. motherboard.vice.com/read/meet-lenna-the-first-lady-of-the-internet-2.

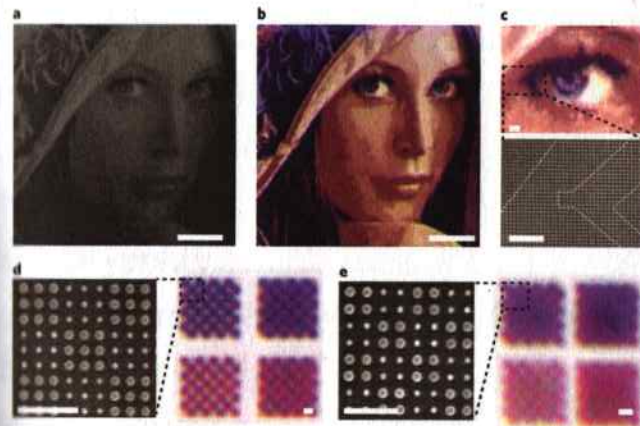
The Face of Image Technology

She is in her 60s, but for image technicians she will be forever 21. 'Lenna', as she is known on the internet, is a legend in the world of image technology. Her image was first used at the University of Southern California at the Signal and Image Processing Institute (SIPI) in 1973. The image technicians were tired of the usual test images. When someone walked in with a magazine, they scanned her image and used it.

This is how Lenna became a favourite amongst image technicians. She is considered a standard for image testing software and online imaging tools, and she even appears in textbooks for technical studies.¹

What is often overlooked is that the portrait is a crop from a 1972 Playboy centrefold of Lena Söderborg, playmate of the month, November. Despite the image's popularity, the use of a soft-erotic image as a technical standard is often cited as an example of sexism in a male dominated computer science world.

On their fiftieth anniversary, the Society for Imaging Science and Technology gave Lenna the title 'First lady of the internet'.²



† Image from: Karthik Kumar, Huigao Duan, Ravi S. Hegde, Samuel C. W. Koh, Jennifer N. Wei. 'Print Colour at the Optical Diffraction Limit'. *Nature Nanotechnology* 7, pp. 557-661, August, 2012.